

CE 310

Week 10 — Multiple Linear Regression

Adding Predictors: Occupancy, Season, and Interaction Effects

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Week 10 Roadmap

Three conceptual moves today — each unlocks a new engineering capability:

Move	Core idea	Engineering payoff
One → many predictors	Hold other variables constant	Isolate the true occupancy premium from temperature
Continuous → categorical	Dummy variables & collinearity	Know when adding a variable hurts more than it helps
Additive → interactive	Slopes that change by group	Size equipment for peak occupied conditions correctly

Data: same 8,760-hour Tucson DOE dataset — adding predictors one at a time.

Goal: interaction model at $R^2 \approx 0.795$, up from 0.601 in Week 9.

Where Week 9 Left Off

Week 9 fitted a single-predictor model: $\hat{y} = 1.1585 \cdot T + (-60.68)$, where T is outdoor temperature in °F.

Result: $R^2 = 0.601$ — outdoor temperature explains **60.1%** of the variance in cooling energy.

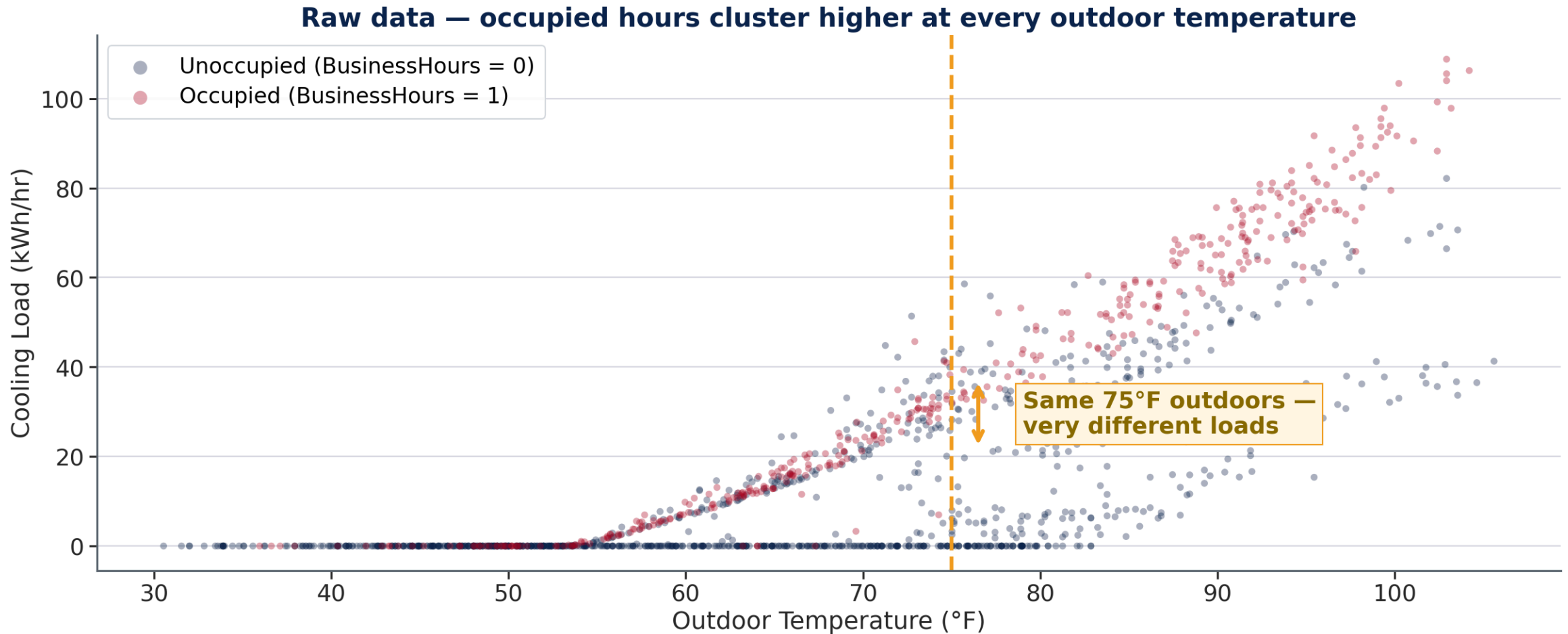
The question for today: **what explains the other 39.9%?**

Consider two hours with the same outdoor temperature — say, 75°F:

Hour	Outdoor Temp	Building	Actual Cooling
Monday 2 pm	75°F	200 people + all computers on	High
Sunday 2 am	75°F	Empty, setback mode	Low

Same temperature. Very different cooling load. Outdoor temperature alone cannot tell these apart.

Two Clouds in the Raw Data



The Missing Predictor: Occupancy

When we split the data by occupancy status:

Group	Mean Outdoor Temp	Mean Cooling (kWh/hr)
Occupied hours	~73°F	38.9 kWh/hr
Unoccupied hours	~64°F	10.6 kWh/hr
Raw difference		28.3 kWh/hr

Occupied hours are both **hotter outside** and **heavier on cooling** — two effects mixed together.

The 28.3 kWh/hr raw difference is not just occupancy — it includes the fact that business hours coincide with the hottest part of the day. Multiple regression separates these two effects.

The Multiple Regression Equation

Add a second predictor alongside outdoor temperature:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

For this dataset:

Symbol	Meaning	Value (Model 2)
x_1	Outdoor temperature (°F)	—
x_2	BusinessHours (1 = occupied, 0 = not)	—
β_1	kWh/hr per 1°F rise, holding occupancy constant	1.026 kWh/°F
β_2	Extra kWh/hr occupied, holding temperature constant	17.9 kWh/hr
β_0	Predicted cooling at 0°F, unoccupied	−55.5 kWh

Fitting MLR in Python

`scipy.stats.linregress` handles one predictor. For multiple predictors, use `statsmodels`:

The model is written the way you would say it aloud — **cooling as a function of outdoor temperature and business hours** — and fitting it returns a summary table with each coefficient, its standard error, and its p -value.

The tutorial covers the syntax and how to read the summary.

Output section	What to read
<code>coef</code> column	$\hat{\beta}$ estimates — one row per predictor
<code>P> t </code> column	p-value: is this coefficient distinguishable from zero?
<code>R-squared</code> / <code>Adj. R-squared</code>	Overall model fit

p-value rule of thumb: $p < 0.05$ means the predictor is real, not noise. Week 11 derives this from first principles.

What "Holding Constant" Means

The phrase *holding the other predictor constant* is the core idea of multiple regression.

Example: $\hat{y} = -56.87 + 1.026 \cdot T + 17.9 \cdot BH$

Prediction at 75°F, occupied (BH = 1):

$$\begin{aligned}\hat{y} &= -56.87 + 1.026(75) + 17.9(1) \\ &= -56.87 + 76.95 + 17.9 = 38.0 \text{ kWh/hr}\end{aligned}$$

Prediction at 75°F, unoccupied (BH = 0):

$$\begin{aligned}\hat{y} &= -56.87 + 1.026(75) + 17.9(0) \\ &= -56.87 + 76.95 + 0 = 20.1 \text{ kWh/hr}\end{aligned}$$

The **17.9 kWh/hr difference** is the occupancy premium — estimated at the **same outdoor temperature**.

This is what Excel LINEST and Week 9's `scipy.stats.linregress` cannot do: they fit one predictor at a time.

Confounding

Raw mean difference: occupied (38.9) – unoccupied (10.6) = **28.3 kWh/hr**

Regression BH coefficient: **17.9 kWh/hr** — where did the other 10.4 kWh go?

Occupied hours coincide with the hottest part of the day:

- Occupied: daytime, weekdays → outdoor temp averages ~73°F
- Unoccupied: nights, weekends → outdoor temp averages ~64°F
- The raw 28.3 kWh gap mixes two effects: occupancy AND the warmer daytime temperatures

Regression holds outdoor temperature constant and isolates the occupancy premium alone. The **17.9 kWh/hr** coefficient is a fairer estimate of what occupancy itself adds.

Part 2: Categorical Predictors & Collinearity

When variables are labels — and when adding one makes things worse

Reading the Output — Coefficients

`statsmodels` produces a detailed output. Here are the coefficient-level rows you need.
(**BH** = BusinessHours indicator: 1 = occupied, 0 = unoccupied)

Row/column	What it tells you	Example (Model 2)
<code>coef</code> column	Estimated slope for each predictor	BH coef = 17.9 kWh/hr
<code>std err</code> column	Uncertainty in the estimate	BH $\approx$ 0.34
<code>t</code> column	Coefficient divided by its std err	BH $\approx$ 52.7
<code>P> t </code> column	p-value: is this coefficient real or noise?	BH $\approx$ 0.000

Rule of thumb: if `P>|t|` < 0.05, the coefficient is statistically significant — the predictor is earning its place in the model.

Reading the Output – Model Fit Statistics

These rows summarise how well the whole model fits.

Row/column	What it tells you	Example (Model 2)
R-squared	Fraction of variance explained	0.697
Adj. R-squared	R ² penalized for extra predictors	0.697
F-statistic	Joint test: are any predictors useful?	large → model is significant

R² vs. Adjusted R²

R² always increases when you add a predictor, even a useless one.

Adjusted R² penalizes for extra predictors. If a new predictor adds less than its share of variance, Adj. R² stays flat or decreases slightly.

Always compare models using **Adjusted R²** when they have different numbers of predictors.

For this dataset ($n = 8760$), the penalty per predictor is tiny — R² and Adj. R² differ by less than 0.001 even in the largest models. The distinction matters more in small datasets or models with many predictors relative to observations.

Quick Check — Does Season Help?

60 seconds · pick A or B

Model 1 (Week 9) used outdoor temperature alone: $R^2 = 0.601$.

Model 2 added BusinessHours: $R^2 = 0.697$.

Now Season is added (Summer / Spring / Fall / Winter) alongside Outdoor_Temp_F.

What do you predict happens to R^2 ?

- **A)** R^2 jumps by another 9+ percentage points — similar gain to BusinessHours
- **B)** R^2 barely moves, staying close to 0.601 — Season adds almost nothing new

✓ Quick Check — Answer: B, Season Barely Helps

R^2 rises from 0.601 to only ~0.611

Model	Predictors	R^2	Gain
Model 1	Outdoor_Temp_F	0.601	—
+ BusinessHours	Outdoor + BH	0.697	+0.096
+ Season	Outdoor + Season	~0.611	+0.010

Why? In Tucson, "Summer" means "outdoor temperature is consistently high." Outdoor temperature already captures this — Season provides almost no new information. This is **collinearity**: two predictors that tell the same story.

💡 Adding a collinear predictor inflates the model, adds complexity, and makes coefficients unstable — without improving prediction.

Collinearity — When Two Predictors Tell the Same Story

Collinearity means two predictors contain overlapping information. Adding a collinear predictor barely changes R^2 .

Season and Outdoor_Temp_F are collinear:

Season	Mean Outdoor Temp (°F)
Summer	~90°F
Spring	~72°F
Fall	~62°F
Winter	~49°F

Collinear predictors also inflate standard errors — it becomes harder to tell which predictor is doing the work.

Detecting Collinearity: VIF

The **Variance Inflation Factor** quantifies how much each coefficient's uncertainty grows because of collinearity.

The **variance inflation factor** is computed for each predictor in turn. VIF above roughly 5 says that predictor is largely explained by the others, and its coefficient should not be read on its own.

The tutorial covers how to compute it.

Feature	VIF	Verdict
Outdoor_Temp_F	3.2	✓ OK — manageable correlation with Season
BH (business hours)	1.4	✓ OK — nearly independent of temperature
Season dummies	4.8–6.1	⚠ Concern — Season and Temp share information

Rule of thumb: $VIF > 5$ = concern · $VIF > 10$ = problem.

Dummy Variables — How a Category Becomes a Number

Regression can only work with numbers. To use a yes/no variable as a predictor, convert it:

A text column of "Yes"/"No" becomes a **0/1 indicator** — 1 when occupied, 0 when not. That single column is what lets a categorical fact enter a numeric model.

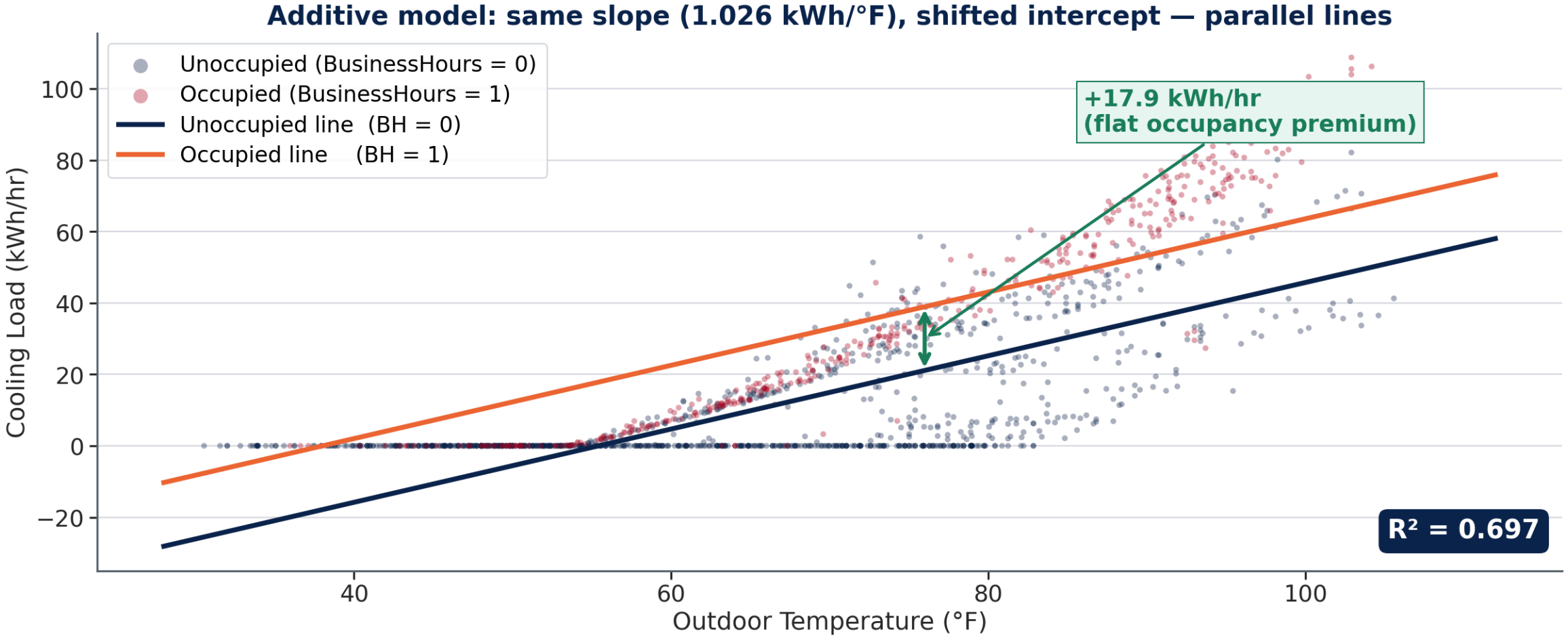
A **dummy variable** is a 0/1 switch used as a *predictor* in linear regression. (In logistic regression a binary variable is the *outcome* — different method, different role.)

In the regression equation:

- When BH = 0: $\hat{y} = \beta_0 + \beta_1 T$ (unoccupied line)
- When BH = 1: $\hat{y} = (\beta_0 + \beta_2) + \beta_1 T$ (occupied line — shifted up by β_2)

Both lines have the **same slope** (β_1). The only difference is the intercept. This is the **additive model** assumption — occupancy shifts the whole line up, but the temperature sensitivity is the same occupied or not.

Additive Model – Two Parallel Lines



Additive Model — What the Parallel Lines Claim

The model assigns the **same slope (1.026 kWh/°F)** to both modes:

- Occupied: $\hat{y} = (\beta_0 + \beta_2) + 1.026 \cdot T = -39.0 + 1.026 \cdot T$
- Unoccupied: $\hat{y} = \beta_0 + 1.026 \cdot T = -56.9 + 1.026 \cdot T$

The occupancy premium is a **flat 17.85 kWh/hr** at every outdoor temperature — the same gap at a cold 40°F night as on a hot 105°F afternoon.

⚠ Is this physically realistic? On a cold night, does occupancy really add as much cooling demand as on a 105°F summer afternoon?

Categorical Predictors with More Than Two Levels

For a variable with k categories (e.g., Season has 4), Python creates $k - 1$ dummy variables. One category becomes the **reference level** (absorbed into the intercept).

Marking a predictor as **categorical** makes the software build the indicator columns for you — one per season, minus the one held back as the baseline.

The tutorial covers the notation.

The output shows coefficients for `C(Season)[T.Spring]`, `C(Season)[T.Summer]`, `C(Season)[T.Winter]` — each is the predicted extra kWh/hr for that season **vs. Fall**, holding outdoor temperature constant.

⚠ When `Outdoor_Temp_F` is already in the model, Season adds almost nothing (R^2 rises from 0.601 to 0.611 only). Outdoor temperature and season are measuring the same underlying phenomenon — how warm it is outside.

Part 3: Interaction Effects & Model Selection

When the slope itself depends on who's in the building

Predict — Same Slope or Different?

60 seconds · discuss with your neighbor

Model 2 says occupancy adds a **flat 17.9 kWh/hr** to cooling at any temperature.

But think physically: on a cold 40°F night, does occupancy add the same cooling demand as on a 105°F summer afternoon?

- **A)** The same slope for both — occupancy is a flat shift, temperature sensitivity is the same
- **B)** Different slopes — occupied buildings are more sensitive to outdoor temperature

✓ Different Slopes — The Interaction Model

Occupancy amplifies temperature sensitivity 2.5×

Mode	Slope (kWh/°F)	Intercept (kWh)	What's driving it
Unoccupied	0.71	−31.2	Envelope conduction only
Occupied	1.78	−78.3	Envelope + people + equipment + lighting

At 105°F:

- Unoccupied: $0.71 \times 105 - 31.2 = \mathbf{43.3 \text{ kWh/hr}}$
- Occupied: $1.78 \times 105 - 78.3 = \mathbf{108.6 \text{ kWh/hr}}$ — 2.5× higher

⚠ The additive Model 2 predicted a flat 17.9 kWh/hr occupancy premium — the interaction model shows it scales from near zero on cold nights to enormous on hot afternoons.

Interaction Terms — When the Slope Changes

The additive model assumes occupancy shifts cooling **up or down by a fixed amount**, regardless of temperature. But is that true?

Intuition: on a 40°F day, the building barely needs cooling whether occupied or not — the occupancy premium is small. On a 105°F day, a fully occupied building adds much more heat from people, computers, and lights on top of an already-hot exterior. The occupancy premium is larger.

This means the **slope itself depends on occupancy**:

- Unoccupied slope: **0.71 kWh/°F** (mostly envelope conduction — heat through walls and windows)
- Occupied slope: **1.78 kWh/°F** (envelope + internal loads from people, lighting, equipment)

The occupied building is **2.5× more sensitive** to outdoor temperature than the unoccupied one.

Interaction Model – Two Equations in One

The interaction model replaces one line with two:

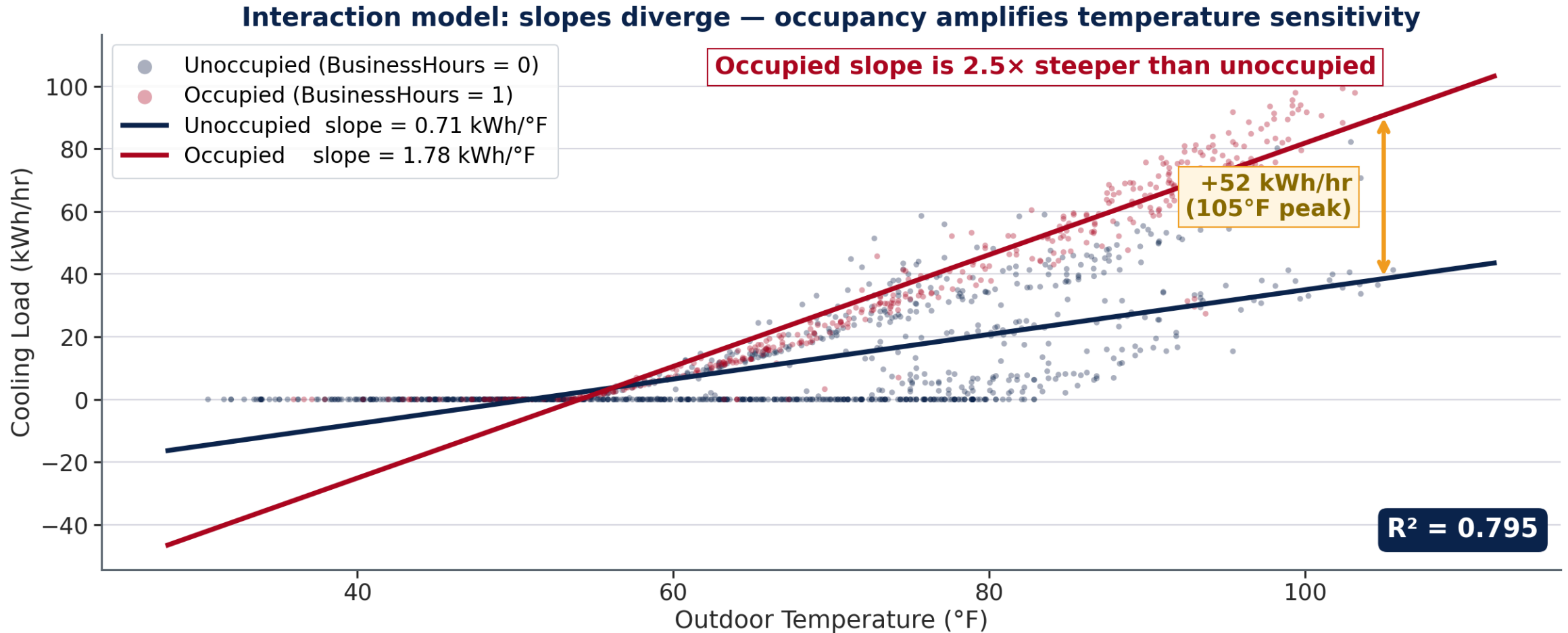
$$\hat{y} = \beta_0 + \beta_1 T + \beta_2 BH + \beta_3 (T \times BH)$$

The term $\beta_3 (T \times BH)$ is the **interaction term**: it adds extra slope when $BH = 1$.

Condition	Equation	Interpretation
Unoccupied ($BH = 0$)	$\hat{y} = \beta_0 + \beta_1 T$	baseline slope = 0.71 kWh/°F
Occupied ($BH = 1$)	$\hat{y} = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)T$	steeper slope = 1.78 kWh/°F

In Python: `smf.ols('Cooling_kWh ~ Outdoor_Temp_F * BH', data=df)` — the `*` automatically creates both main effects and the interaction term.

Interaction Model – Two Diverging Lines



Interaction Model — What the Gap Means

The occupied line is **2.50× steeper** than the unoccupied line. At 105°F the occupied building demands ~52 kWh/hr more than the unoccupied one — a gap that grows with temperature, not a flat shift. The additive model misses this entirely.

Which Model Wins?

60 seconds · pick A, B, or C

Three models, all using the same dataset. Two metrics reported:

Model	Predictors	k	R ²	AIC
A	Outdoor_Temp_F only	2	0.601	73,407
B	Outdoor_Temp_F + BH + Season	6	0.702	70,854
C	Outdoor_Temp_F × BH (interaction)	4	0.795	67,571

Which model would you select for an engineering report?

- **A)** Model A — simplest, fewest assumptions
- **B)** Model B — more predictors captured, higher R²
- **C)** Model C — best R² and lowest AIC with only 4 parameters

✓ Model C — Best on Both Metrics

Higher R^2 , lower AIC, and fewer parameters than Model B

Model	Predictors	k	R^2	AIC	Δ -AIC	Verdict
A	Temp only	2	0.601	73,407	+5,836	Baseline
B	Temp + BH + Season	6	0.702	70,854	+3,283	Season adds complexity without payoff
C	Temp $\times$ BH	4	0.795	67,571	0	✓ Wins on every criterion

Model B uses 6 coefficients for $R^2=0.702$. Model C uses 4 coefficients for $R^2=0.795$. The interaction term extracts more information than Season does — with fewer parameters.

Δ -AIC > 10 is decisive. Model C beats B by 3,283 — a commanding margin.

Model Comparison — R² Progression

Each model adds one idea. The R² gains tell you whether that idea was worth adding.

Model	Predictors	R ²	What it added
Week 9 simple	Outdoor_Temp_F	0.601	Temperature baseline
Model 2	+ BusinessHours	0.697	Occupancy: +9.6 pp
Model 3	+ Season	0.611	Season: only +1.0 pp (collinear)
Model 4	+ BH + Season	0.702	Season adds 0.5 pp on top of BH
Model 5	Temp × BH interaction	0.795	Slopes differ: +9.8 pp

The interaction model (Model 5) is clearly the best fit. But it comes at the cost of a more complex equation with four coefficients instead of three.

Model Selection — AIC

AIC (Akaike Information Criterion): lower = better. Every extra coefficient costs a penalty — even when R^2 increases.

Model	Predictors	k	R^2	AIC	Δ -AIC	Verdict
Model 1	Temp	2	0.601	73,407	+5,836	Baseline
Model 2	Temp + BH	3	0.697	70,992	+3,421	BH earns its place
Model 3	Temp + Season	5	0.611	73,181	+5,610	⚠️ 3 extra params, tiny R^2 gain
Model 4	Temp + BH + Season	6	0.702	70,854	+3,283	Season marginal on top of BH
Model 5	Temp × BH	4	0.795	67,571	0	✅ Best R^2 and lowest AIC

Δ -AIC > 10 is decisive. Model 5 beats Model 2 by 3,421 — not close. Model 3 (Season) uses 5 parameters for a 0.010 R^2 gain; its AIC barely improves vs. Model 1. The interaction term (Model 5) extracts 0.098 more R^2 than Model 2 using only one additional parameter.

Residuals — Did Adding Predictors Help?

The residual plot is the best diagnostic for model fit.

Week 9 simple model

- Clear fan shape: residuals spread wider at high temperatures
- Cluster of positive residuals at $T < 45^{\circ}\text{F}$ (chiller off, model predicts negative cooling)
- Pattern = the model is systematically wrong in two temperature ranges

Model 5 (interaction)

- Fan shape mostly gone — occupied and unoccupied hours now follow separate lines
- Low-temperature cluster reduced — separate intercepts let unoccupied fit lower
- Still some structure, but much closer to random scatter

The goal is residuals that look like **random noise** — no trend, no fan shape, no clusters. Perfect randomness means the model has captured all systematic variation.

Part 4: Connections

MLR across CE disciplines and into machine learning

MLR Across CE Domains

Multiple regression with categorical and interaction predictors spans every CE subfield:

Structural Engineering

- Predictors: dead load + live load + wind (each independent)
- Interaction: wind $\times$ height (taller buildings amplify more)

Water Resources

- Predictors: rainfall + soil moisture + season dummy
- Season dummy: wet/dry season shifts turbidity response

Transportation

- Speed $\sim$ volume + truck% + rain dummy (shifts speed-flow curve)
- VIF check: rush-hour dummy vs. volume often collinear

Energy Systems

- EUI $\sim$ HDD + CDD + occupancy type (k-level) + vintage dummy
- Vintage $\times$ CDD: older buildings react more sharply to heat

Bridge to Machine Learning

The ideas from today are the foundation of how ML models work.

Statistics term	ML equivalent	Connection
Multiple regression	Linear model / ridge / lasso	Same equation, different solver
Adding predictors	Feature engineering	Selecting which x-variables to use
Dummy variable	One-hot encoding	Same concept, standard preprocessing
Interaction term ($T \times BH$)	Feature cross	Feature product used in boosting
Adjusted R^2	Validation score / RMSE	Both penalize overfitting

The **interaction model** is exactly how tree-based models (random forests, gradient boosting) work: each split says "if $BH=1$, use this slope; if $BH=0$, use that slope." Today's four-coefficient model is the simplest possible version of that idea.

Python Quick Reference — Week 10

Concept	Python
Fit multiple regression	<code>smf.ols('y ~ x1 + x2', data=df).fit()</code>
Create dummy variable	<code>df['BH'] = (df['BH']=='Yes').astype(int)</code>
Auto-dummies (k levels)	<code>C(Season)</code> inside formula string
Interaction term	<code>smf.ols('y ~ x1 * x2', data=df).fit()</code>
Check VIF	<code>variance_inflation_factor(X.values, i)</code>
Compare models	<code>.aic</code> attribute on fitted model

Model Selection Checklist

Apply in order when building any multi-predictor model:

1. **Start** with physically motivated predictors
2. **Check VIF** — remove or combine if $VIF > 5$
3. **Add interaction** if physical reasoning supports different slopes by group
4. **Compare** Adj. R^2 and AIC — always plot residuals before reporting

Looking Ahead — Week 11: Hypothesis Testing

Today you used p-values to test whether each coefficient is real. Week 11 formalizes this idea from first principles.

The setup: occupied hours average 38.9 kWh/hr; unoccupied average 10.6 kWh/hr. Is the 28.3 kWh gap real, or could it be sampling noise?

Week 11 tools:

- **Two-sample t-test:** tests whether two group means differ
- **H_0 :** the difference is zero — the groups are the same
- **p-value:** probability of seeing this gap if H_0 were true
- **Connection:** the BH p-value in today's regression is doing this same test